WHITEPAPER

OPTIMIZED SEALING FOR THERMAL AND CHEMICAL RESISTANCE

SIMRIZ® FFKM COMBINES THE BEST OF TRADITIONAL PEROXIDE AND TRIAZINE CURING SYSTEMS FOR IMPROVED OVERALL PERFORMANCE

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SEALING TECHNOLOGIES

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Initial Situation

Perfluoroelastomer (FFKM) compounds provide excellent sealing characteristics in O-rings, gaskets and other components across a wide variety of applications and industries. Containing even more fluorine than FKM, FFKM elastomers are used in seals that involve contact with hydrocarbons, highly corrosive fluids and/or extreme temperatures.

Traditionally, two main curing systems have been used – peroxide or triazine – depending on the operating conditions to which the end product will be subjected. Peroxide-based cures excel when it comes to chemical resistance (including strong acids); triazine cures provide increased thermal stability.

In recent years, however, conditions have become even more severe. Customer equipment and their corresponding seals are often exposed to both harsh chemicals and extended high temperatures in demanding applications, such as jet engines, semiconductor fabrication equipment and oil well blowout preventers. These can be dangerous and safety-critical applications, where failure is not an option. At the same time, customers expect products to last longer, while maintaining short delivery lead times, global support and affordable pricing.

As a result, users have had to make compromises, sacrificing some chemical or temperature resistance properties for the other by choosing either a peroxide- or triazine-based curing system. This limits overall performance and results in parts having to be replaced more frequently.

The Freudenberg Solution

Freudenberg Sealing Technologies has developed a proprietary curing system that combines the benefits of traditional peroxide and triazine methods, while reducing costs by as much as 50 percent. Marketed under the company's Simriz® brand, the high-performance compounds feature a patented triazine crosslink with polymer structure modifications.

Simriz® possesses the resilience of an elastomer with the chemical resistance approaching that of a polytetrafluoroethylene. Seals using Freudenberg's proprietary curing technology can withstand temperatures as high as high as 325 °C (617 °F), while providing extremely broad chemical resistance against amines and strong acids – as well as outstanding performance under steam and hot water exposure – with unmatched compression-set performance. In short, the aromatic thermally stable crosslink design offers the best of both traditional curing systems in a single high-performance compound.

The Simriz® family ranges from traditional peroxide-based compounds to several proprietary grades to meet customer-specific requirements. The latter group consists of:

Simriz®498
The ultimate FFKM material for applications in the chemical processing and petroleum industries.

Simriz®501
Designed for aerospace applications.

Simriz®502
A stiffer 90-durometer design to protect against high pressure and decompression in oil and gas seals.

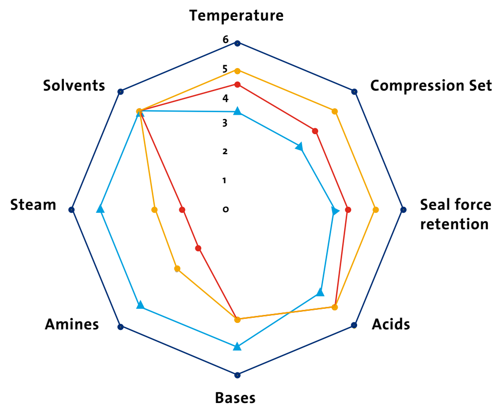
Simriz®508
An ultra-clean, low-particulate white material (the previous three are carbon black) designed for the semiconductor industry.

Verified Performance in Extreme Testing and Years of Customer Use

Freudenberg's proprietary compounds are far superior to the top grades of its competitors. The results have been verified during testing under extreme conditions. After being subjected to 168 hours at 230°C (446°F) in steam, the Simriz® 498 O-rings came out looking like new. The competition didn't fare nearly as well, with the products of two main rivals suffering degradation, swelling and nucleophilic attack on their crosslinking systems.

Simriz® seals also have accumulated more than 15 years of data and real-world experience in some of the most demanding O-ring applications imaginable. This includes down hole/SagD blowout preventers in the oil and gas industry, automated paint spray equipment in automotive factories and air management "bleed air systems" on jet plane engines. Maintaining sealing performance in these functions is crucial. To this end, the superior capabilities of the proprietary curing system helped Simriz® seals displace both the original equipment and aftermarket O-ring suppliers of one engine manufacturer.

5=Competitor Materials best performance
6=Simriz® 498 performance



Competitor Material 1 (Strong Chemical Resistance)
Competitor Material 2 (General Purpose)
Competitor Material 3 (High Temperature)

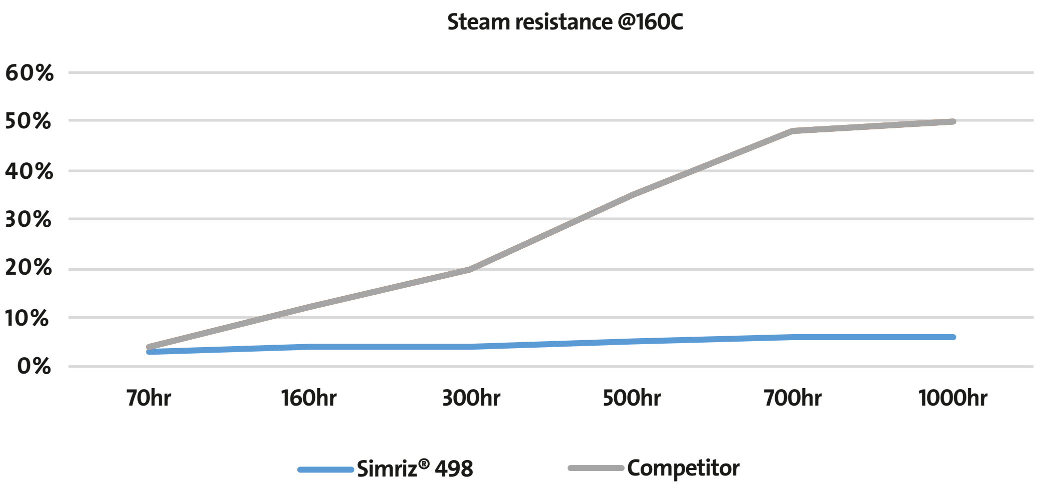
Vertically Integrated, No-Compromise Cost-Effective Solution

The benefits add up quickly with Simriz®. Freudenberg is the only fully integrated O-ring manufacturer, supplying base monomers and polymers to the additives and final compounds. Not only does this cut out the middlemen, but it also results in significant cost savings, faster development times, greater design flexibility and less complexity for suppliers.

From a technical standpoint, Freudenberg's proprietary curing system provides unparalleled simultaneous chemical and thermal resistance. O-rings, gaskets and other products made from these compounds are designed to meet the most demanding industry challenges without any compromises.

YOUR BENEFITS AT A GLANCE

- Patented crosslink technology
  – Proprietary design combines the benefits of traditional triazine- and peroxide-based curing systems
  – A no-compromise solution for applications with both harsh chemical and extreme temperature requirements
- Chemical resistance approaching PTFE
  – Amines
  – Bases
  – Solvents
  – Strong acids
- Extreme thermal stability
  – Can withstand temperatures up to 325 °C (617 °F)
  – Outstanding performance in steam and hot water
- Cost-efficient
  – Reduced manufacturing costs
  – Shorter lead times
- Vertically integrated
  – Freudenberg is the only O-ring manufacturer vertically integrated back to monomers and polymers



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About Freudenberg Sealing Technologies
Freudenberg Sealing Technologies is a company with many years of technological expertise and a global market leader for demanding and innovative applications in sealing technology and e-mobility. Thanks to its unique materials and technology expertise, the company is a proven supplier of advanced products and applications as well as a development and service partner for customers in the automotive industry and general industry. In fiscal year 2020, Freudenberg Sealing Technologies generated sales of some EUR 1.9 billion and employed approximately 13,000 people. Further information can be found at www.fst.com

The company is part of the globally active Freudenberg Group, which generated sales of some EUR 8.9 billion with its Seals and Vibration Control Technology, Nonwovens and Filtration, Household Products and Specialties and Others Business Fields in fiscal year 2020 and employed approximately 48,000 people in around 60 countries. Further information at www.freudenberg.com

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